
LLM Anonymization Against Agentic Re-Identification

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Abstract

Agentic LLMs with web search change the threat model for text anonymization: weak contextual cues can become cross-referenceable evidence for re-identification, yet those same details also carry downstream analytic value of the text. Existing defenses either remove explicit identifiers, perturb text for formal privacy, or test rewritten text against non-web inference models, leaving underexplored the operating region between resistance to agentic web-search re-identification and utility retention. We introduce AURA (Anonymization with Utility-Retention Adaptation), an LLM-powered *mask-reconstruct* framework that decouples privacy localization from utility-preserving reconstruction and selects candidates with adversarial privacy and utility-retention checks. We evaluate AURA on real-user interview transcripts using re-identification attacks carried out by web-search agents, along with a utility evaluation based on interviewee-profile facts, codebook facts, and the joint contextual utility grid. Our results show that AURA improves the privacy-utility frontier by using adaptive privacy scope to strengthen resistance to agentic re-identification and using a mask-reconstruct anonymization method to better preserve contextual utility under fixed privacy scope¹.

1 Introduction

The rapid growth in large language models (LLMs) [1, 39] capabilities and adoption has renewed debate about their societal impact, with privacy emerging as a central concern beyond early work on training-data memorization. Recent works have shown that modern agentic LLM systems can use web search, generate queries from weak contextual cues, retrieve public evidence, and cross-reference external materials to infer identities [34, 19, 18, 16]. This makes LLM-era anonymization less a problem of deleting explicit identifiers than of deciding which contextual details can safely remain.

The stakes are amplified by the growing wave of large-scale AI use data collection and re-distribution. Public datasets such as LMSYS-Chat-1M [47], WildChat [46], SWE-chat [4], Anthropic Interviewer [11], release increasingly rich records of how people use AI systems, while providers conduct proprietary analyses that require the same richness (e.g., Anthropic’s Clio [38], the Anthropic Economic Index [2], and OpenAI’s NBER study [6]). These efforts share a common requirement:

¹Source Code: <https://github.com/AaronLi43/AURA>

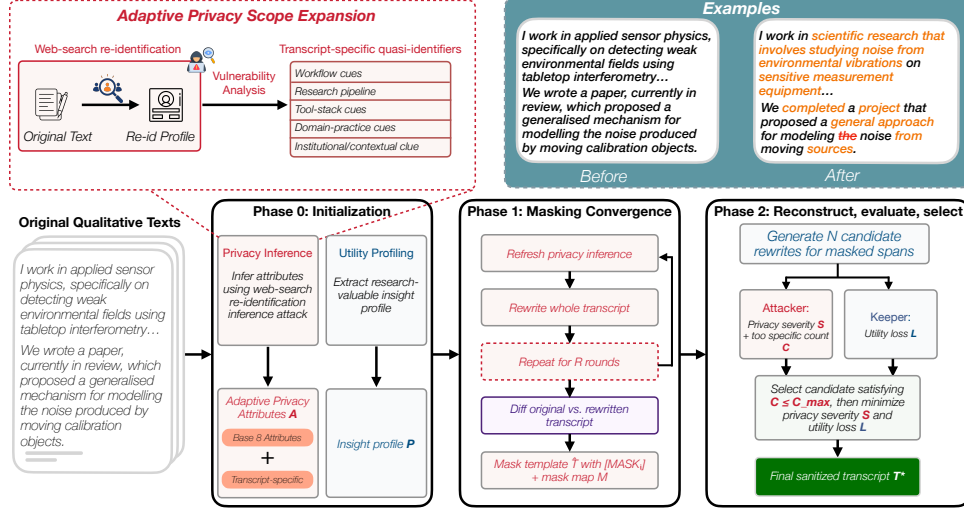


Figure 1: AURA overview. Adaptive privacy scope expansion first augments a base re-identification profile with transcript-specific quasi-identifiers, then AURA initializes privacy and utility profiles, converges on a mask template for sensitive spans, and reconstructs only those spans based on the masked original transcript. Candidate rewrites are evaluated by an attribute inference attacker and a keeper before selecting the final sanitized transcript T^* .

research questions that cut across respondent background, behavioral patterns, domain expertise, and attitudinal reasoning demand data that retains contextual nuance across multiple dimensions simultaneously. Yet the same contextual details that carry analytic value are precisely the cues that agentic LLMs exploit for re-identification [19, 18, 16], creating a structural tension between the privacy of participants and the utility of the data they provide.

Current anonymization methods are inadequate for this challenge. Heuristic sanitization methods represent the status quo in real-world data sanitization practices. In particular, Named-entity recognition (NER) systems such as Microsoft Presidio [26] detect explicit identifiers (names, emails, dates) but miss the contextual inference cues that modern LLMs exploit [34]. LLMs demonstrate potential to advance anonymization by moving beyond explicit identifier redaction to hiding sensitive attributes from model inferences [35]. At the formal end of the spectrum, differential privacy is the predominant framework for privacy guarantees in machine learning, but DP text rewriting methods [24, 23, 41] often achieve these guarantees by perturbing text at the token level, which can substantially degrade readability and analytical value. Practical DP deployment requires theoretically managing privacy-utility tradeoffs using privacy budget parameters as tuning knobs [30, 10, 17] and investigating the impact on privacy and utility empirically [13].

We ask this question: *How can we anonymize text against the pragmatic agentic re-identification threats while preserving the information needed for downstream use?* It boils down to two gaps. **First**, agentic attacks make sensitive span localization *beyond predefined categories* necessary because the attack success is contingent on the availability of contextual cues which not only include typical personal attributes but also reflect niche, idiosyncratic details that only become identifying when combined with external evidence. **Second**, identifying these cues is not enough. Unlike explicit identifiers, contextual cues that create re-identification risk often carry substantive downstream insights. Anonymization must therefore decide not only which spans are risky but how to transform them to preserve utility.

Thus, we introduce **AURA** (Anonymization with **U**tility-**R**etention **A**daptation), an LLM-powered *mask-reconstruct* framework for LLM-era text anonymization. AURA separates privacy localization from utility-preserving reconstruction: it first identifies contextual cues that may support agentic web-search re-identification, then reconstructs the affected spans under empirical constraints and metrics of privacy and utility. We evaluate AURA and other anonymization methods on real-world interview transcripts from the Anthropic Interviewer dataset [11] verified as vulnerable to agentic

re-identification. We applied AURA variants under closed-source and open-weight LLM backbones, along with five other anonymization methods to these transcripts and then tested the outputs against agentic re-identification attacks. Across three deanonymization attacker models (GPT-5.4-mini, GPT-5.1, and Gemini-3-Flash), AURA’s adaptive-privacy variants reduce agentic re-identification to 0-5/27 transcripts, substantially below NER-based redaction (13-21/27) and the prior LLM-based anonymizer [35] (6-7/27) while retaining 74.9-80.3% of unit-level utility-grid information, outperforming baselines on the privacy-utility frontier. On-device backbones such as Qwen3.5-27B and Qwen3.5-35B-A3B match or exceed the API-powered baseline on utility (78.7%/80.2% vs. 72.1% unit-grid recovery) at comparable privacy, supporting fully local deployment.

Our main contributions are:

1. To our knowledge, we are the first to optimize and evaluate LLM text anonymization in the operating region between resistance to real-world agentic web-search re-identification and retention of downstream analytic utility.
2. We propose **AURA**, an LLM-powered *mask-reconstruct* framework that decouples *where* to intervene from *how* to rewrite, and is evaluated with both agentic re-identification attacks and utility-retention checks.
3. We empirically characterize how privacy and utility change across anonymization settings, showing that attribute scope primarily influences resistance to re-identification while reconstruction preserves utility while maintaining privacy.

2 Related Work

Text de-identification has long focused on the removal of explicit identifiers based on predefined taxonomies. The related technical problem is named entity recognition (NER), which has progressed from rule-based systems to neural models that approach human performance on standard benchmarks [25, 36, 8, 29]. Publicly available NER tools such as Presidio [26] have been widely cited in research to support anonymization in textual data release [47, 46, 4, 20]. However, releasing qualitative data containing detailed behavioral or attitudinal textual data requires higher standards for anonymization because interview transcripts contain free-form disclosures whose identifying power emerges from context and attribute combinations [28, 22, 31], which named-entity removal alone cannot address. This type of data is usually treated with high caution and traditionally based on manual anonymization [32, 37, 12, 5], which remains ad-hoc and impractical to scale, and can especially become vulnerable to intensified real-world re-identification threats that are democratized and scalable by LLMs [19, 16, 18]. AURA directly tackles this gap by localizing and transforming contextual quasi-identifiers in long-form transcripts rather than only deleting named entities.

Recent work has begun to use LLMs for text sanitization and anonymization, some explicitly modeling privacy-utility trade-offs [43, 33, 48, 35]. However, these works operationalize privacy and utility in qualitatively different ways. Some address privacy leakage to remote LLM providers by redacting [33] or abstracting [33, 48] NER-based PII before prompts are shared. A second, more nascent line of work studies inferential privacy risks [27], proposing defenses that include iteratively rewriting text to suppress personal attributes from being inferable [35] or prevent non-agentic LLM-based re-identification [43]. Our work targets a distinct and stronger threat model: agentic re-identification, where an LLM agent combines textual cues with web search evidence to identify ordinary individuals with linkable online traces. This setting is both more dynamic and more costly to defend against, making it impractical to place an agentic attacker inside every rewriting iteration. AURA’s mask-reconstruct framework provides a solution by decoupling the problem into two parts: 1) a one-off run of an agentic attacker to identify attributes that can be used to re-identify the interviewee; 2) an iterative rewriting process that only uses LLM-based attribute inference to check the preservation of the attributes resulting from the first step. To establish a direct comparison, we included Staab et al. [35] and two one-shot LLM rewriting settings as baselines.

At the formal end of the design space, differential privacy (DP) rewrites text via synthetic representations, word-level noise, or DP-fine-tuned generators [9, 42, 23, 15, 41, 24, 7, 44, 45, 3]. However, strong perturbation often damages coherence and analytic value in long-form qualitative text, and privatized text can still face reconstruction attacks [40], leaving a gap between readable but leaky rewrites and private but low-utility outputs. We evaluated DP-based text rewriting methods as baselines to probe their empirical privacy-utility tradeoffs against agentic re-identification.

Algorithm 1 AURA mask–reconstruct pipeline

Require: Original transcript T ; phase 1 masking rounds R_{mask} ; phase 2 candidate batch size N

Ensure: Final sanitized transcript T^* ; masked template $\hat{T}$; mask map $M = \{i \mapsto s_i\}$; selected replacement dictionary R^*

Phase 0: Initialize privacy and utility context

- 1: Generate privacy scope $\mathcal{A}$;
- 2: Infer initial privacy attributes $\Pi^{(0)}$ and evidence spans B from T using $\mathcal{A}$
- 3: Extract utility/insight profile $\mathcal{P}$ from T

Phase 1: Converge on risky spans to mask

- 4: Initialize working transcript $T^{(0)} \leftarrow T$
- 5: **for** $r = 1$ to R_{mask} **do**
- 6: Refresh privacy inferences $\Pi^{(r)}$ on $T^{(r-1)}$. Break when no attribute inferred
- 7: Rewrite $T^{(r-1)}$ conditioned on $\Pi^{(r)}$ to reduce attribute leakage, yielding $T^{(r)}$
- 8: **end for**
- 9: Diff original transcript T against converged rewrite $T^{(R_{\text{mask}})}$
- 10: Construct masked template $\hat{T}$, mask map $M = \{i \mapsto s_i\}$ from mask IDs to original spans, and seed replacements S_{seed}

Phase 2: Reconstruct masked spans and select a candidate

- 11: Generate N replacement dictionaries $\{R^{(1)}, \dots, R^{(N)}\}$
 - 12: Assemble candidate rewrites $\{T^{(1)}, \dots, T^{(N)}\}$
 - 13: **for all** $T^{(n)}$ in parallel **do**
 - 14: Run attribute inference attacker on $T^{(n)}$ with privacy scope $\mathcal{A}$
 - 15: Run utility keeper on $(T, T^{(n)}, M, \mathcal{P})$ to score retained utility
 - 16: Compute specificity count C_n , privacy severity S_n , and utility loss L_n
 - 17: **end for**
 - 18: Let $\mathcal{V} = \{n \mid C_n \leq C_{\text{max}}\}$ denote candidates satisfying the specificity cap
 - 19: **if** $\mathcal{V} \neq \emptyset$ **then**
 - 20: $n^* \leftarrow \arg \min_{n \in \mathcal{V}} (S_n, L_n)$
 - 21: **else**
 - 22: $n^* \leftarrow \arg \min_n (C_n, S_n, L_n)$
 - 23: **end if**
 - 24: **return** $T^* \leftarrow T^{(n^*)}$, $\hat{T}$, M , and $R^* \leftarrow R^{(n^*)}$
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3 Method

We present AURA as a two-phase decomposition of text anonymization that balances privacy and utility preservation. Rather than asking one model to generically rewrite an entire interview, AURA first localizes privacy-bearing spans through masking, generates a batch of reconstructed spans to fill the blanks, and selects the final sanitized transcript via adversarial privacy and utility-retention check. The pipeline described in Algorithm 1 can be summarized as: (i) Phase 0: Initialization, (ii) Phase 1: Masking Convergence, and (iii) Phase 2: Reconstruct, Evaluate, and Select. Appendix B shows all the LLM prompts used in the three phases.

3.1 Phase 0: Initialization

We use LLMs with web search to initialize a privacy scope $\mathcal{A}$ for the given interview transcript T . Based upon a predefined base privacy scope containing eight attribute types from [35]: *Age*, *Sex*, *Location*, *Occupation*, *Education*, *Relationship Status*, *Income*, and *Place of Birth*, we use the LLM to determine additional personal attributes that could be used to re-identify the interviewee and add them to the final privacy scope, as well as the evidence spans in the transcript which form a blacklist B . An insight profile $\mathcal{P}$ summarizing the transcript’s topic is also extracted across eight utility dimensions (*Thematic Content*, *Experiential Narratives*, *Emotional/Affective Expressions*, *Reasoning & Beliefs*, *Behavioral Patterns*, *Relational Dynamics*, *Temporal Structure*, and *Domain Knowledge*).

AURA variants. In addition to the main AURA design that uses an adaptive privacy scope (**adaptive privacy AURA**), we also included variants for ablation evaluation. The **8-attribute AURA** variant

has the same privacy scope as the anonymizer baseline [35]. We also tested a **pure adaptive AURA variant**, which directly infers identifiable personal attributes without the 8 attributes as the base set.

3.2 Phase 1: Masking Convergence

Phase 1 runs an iterative rewriting process with privacy-inference feedback from LLMs. Starting from an initial text $t_0 = T$, each iteration i takes the current text t_i as input, uses an LLM to infer attributes in the privacy scope A , and then produces a rewritten text t_{i+1} with (t_i, A_i) , repeating until no attributes can be inferred or a stopping condition such as hitting masker convergence rounds is met. We generate masks using a diff between the original and rewritten text, and using the final output text at this phase to derive the seed replacement of each masked span for Phase 2.

The masker’s outputs are: a masked template $\hat{T}$ with placeholders $[\text{MASK}_i]$, a mask map $\{i \mapsto s_i\}$ from placeholders to original spans, and the seed replacements that are the spans under the masker before reconstruction. If no masks are produced, the pipeline returns the original transcript unchanged.

3.3 Phase 2: Reconstruct, Evaluate, and Select

Given the masked template $\hat{T}$, the original mask map, and seed replacements from Phase 1, as well as the insight profile from Phase 0, the reconstructor generates N candidate replacement dictionaries only for the masked spans, rather than rewriting the full transcript. Although the original masked spans are fed in the mask map as references, the reconstructor is instructed not to copy them verbatim and instead to reconstruct safe replacements from the masked context and insight profile. The N replacement dictionaries can be used to generate candidate rewrites $\{T^{(1)}, \dots, T^{(N)}\}$. The final result is selected from the N candidate rewrites by evaluating privacy, utility, and specificity. Hyperparameter settings are included in Appendix A. Appendix B.3 shows the prompts of reconstructors.

Candidate Selection. Each candidate rewrite $T^{(n)}$ is evaluated by three scores. First, the attribute inference attacker re-runs privacy inference on the rewritten transcript using the privacy scope $\mathcal{A}$, compares the inferred attributes with the Phase 0 privacy inferences, and sums the resulting attribute-level leakage severities into a privacy score $S_n = \sum_a \text{severity}_{n,a}$. Second, the attacker flags dimensions on the specificity checklist C that remain overly specific and counts them as C_n , where lower C_n means the rewrite better satisfies the desired generalization level. A list of 5 dimensions designed for [11] is included in the prompt of Specificity Auditor in Appendix B.4. Customized C may be needed for different datasets. Third, the keeper compares the original transcript, the candidate rewrite, and the mask map to score how much research-valuable content was lost across the eight utility dimensions and sums them up, yielding $L_n = \sum_u \text{loss}_{n,u}$.

Candidate selection is privacy-first. AURA first filters to the admissible set $\mathcal{V} = \{n : C_n \leq C_{\max}\}$ where $C_{\max}$ is the maximum number of dimensions identified as too specific, enforcing that the rewrite is not too specific on more than the allowed number of attributes. If $\mathcal{V}$ is non-empty, AURA selects the candidate with the lowest privacy severity S_n and uses lower utility loss L_n only as a tie-breaker. If no candidate satisfies the specificity cap, AURA falls back to the best available candidate by minimizing (C_n, S_n, L_n) in order, so specificity is repaired before severity and utility.

4 Experimental Setup

We develop the benchmarks for privacy preservation and utility retention based on 27 re-identifiable interview transcripts from the Anthropic Interviewer dataset [11]. To construct it, we apply the agentic re-identification attack from prior work [19] to all 1,250 transcripts and retain only cases with verifiable identification evidence, i.e., a specific individual or small set of individuals (e.g., paper authors) have been identified as the interviewee. The resulting set is small yet challenging, including long, information-rich interview transcripts from real individuals with validated re-identification risks, making it suitable for evaluating anonymization methods.

4.1 Privacy Benchmark: Agentic re-identification

We evaluate privacy by measuring whether an agentic LLM with web-search access can re-identify the interviewee from the rewritten transcript. We report both the count and percentage of transcripts

re-identified under three attacker models: GPT-5.1², GPT-5.4-mini³, and Gemini-3-Flash⁴ to mitigate the model bias. The same attack protocol is used for privacy scope expansion. Appendix B.7 shows the prompt for web-search re-identification and privacy scope expansion. Cross-attacker robustness will be discussed in later sections. Additionally, we provide a synthetic qualitative diff analysis comparing AURA’s rewrites against the advanced anonymizer baseline [35] to examine the scope of edits (Appendix E).

4.2 Utility Benchmark: Profile, Codebook, and Utility-Grid Unit Recovery.

We mirror real-world analysis pattern in our utility benchmark. In practice, researchers often interpret what a participant says together with who that participant is, their occupational context, and their behavioral patterning [11, 14]. This follows the qualitative research emphasis on thick description, where preserving contextualized accounts of participants’ experiences supports the transferability of findings [21]. We therefore operationalize utility at three levels: 1) interviewee-profile facts, 2) behavior-codebook facts, and 3) their combinations in a utility grid that approximates downstream qualitative question formation [14]. After filtering to facts validated on the original transcripts, the combined utility-grid benchmark contains 170 profile facts, 371 code facts, and 2,349 profile-code units over the 27 transcripts. Appendix F presents the utility benchmark construction methodological details, including the human expert coding process, the codebook, fact recovery as utility metrics and LLM-as-a-judge setting and prompts, and representative utility-grid unit examples.

4.3 Baselines and AURA Variants

We compare the AURA variants with different privacy scopes against external baselines spanning NER-based de-identification, end-to-end LLM rewriting, iterative adversarial LLM anonymization, and differentially private rewriting. Full baseline descriptions are provided in Appendix D.

1. **Presidio** [26]: Microsoft’s open-source NER-based PII detection and replacement tool.
2. **Minimal one-shot rewriting**: End-to-end LLM rewriting with a minimal instruction to simulate the day-to-day usage of an anonymizer.
3. **Detailed one-shot rewriting**: End-to-end LLM rewriting with a detailed prompt specifying what to change and what to preserve, emulating the goals of our pipeline in a single pass.
4. **Advanced Anonymizer** [35]: Iterative adversarial LLM anonymization with feedback guides.
5. **DP-MLM** ($\epsilon \in \{10, 30, 50, 70, 100, 120, 140\}$) [24]: Differentially private text rewriting using masked language models with per-token ϵ -DP guarantees.

We included AURA variants using proprietary models (GPT-5.1 for re-identification-based privacy scope expansion and GPT-4.1 for all other tasks) and open-weight models: Deepseek-V4-Flash⁵ with web-search tools for privacy scope expansion, and two Qwen models for other tasks.

5 Results

5.1 Privacy Analysis

Agentic re-identification. Table 1 reports results under three attacker models. DP-MLM is the most privacy-preserving baseline at low ϵ (0/27 re-identifications at $\epsilon=10, 30$ across all attackers), while its looser settings become modestly re-identifiable (1–5/27 at $\epsilon \geq 50$). Among non-DP baselines, the adaptive AURA variants are consistently the least re-identifiable across all three attackers (0–5/27), indicating that transcript-specific privacy scope effectively reduced linkage risk. The fixed-scope methods cluster together across attackers (8-attribute AURA at 2–8/27 and the anonymizer at 6–7/27), whereas Presidio (13–21/27) and one-shot rewriting (8–17/27) remain substantially more at risk.

²<https://openai.com/index/gpt-5-1/>

³<https://openai.com/index/introducing-gpt-5-4-mini-and-nano/>

⁴<https://docs.cloud.google.com/vertex-ai/generative-ai/docs/models/gemini/3-flash>

⁵<https://huggingface.co/deepseek-ai/DeepSeek-V4-Flash>

Table 1: Agentic re-identification on 27 transcripts using GPT-5.1, GPT-5.4-mini, and Gemini-3-Flash as attacker models. We repeated each attack three times and report the highest re-identification rate. The methods showing the best privacy-utility tradeoff are bolded. No model refusal was observed.

Method	GPT-5.1		GPT-5.4-mini		Gemini-3-Flash	
	Re-ID	Rate	Re-ID	Rate	Re-ID	Rate
AURA (adapt. privacy, Qwen3.5-27B)	2/27	7.4%	4/27	14.8%	0/27	0.0%
AURA (adapt. privacy, Qwen3.5-35B-A3B)	2/27	7.4%	5/27	18.5%	2/27	7.4%
AURA (adapt. privacy, GPT-4.1)	2/27	7.4%	3/27	11.1%	0/27	0.0%
AURA (pure adaptive, GPT-4.1)	2/27	7.4%	3/27	11.1%	2/27	7.4%
AURA (8-attribute, GPT-4.1)	6/27	22.2%	8/27	29.6%	7/27	25.9%
AURA (8-attribute, Qwen3.5-27B)	4/27	14.8%	7/27	25.9%	3/27	11.1%
AURA (8-attribute, Qwen3.5-35B-A3B)	2/27	7.4%	8/27	29.6%	4/27	14.8%
Anonymizer	6/27	22.2%	7/27	25.9%	7/27	25.9%
Presidio	13/27	48.1%	21/27	77.8%	17/27	63.0%
One-shot w/ minimal prompt	10/27	37.0%	14/27	51.9%	8/27	29.6%
One-shot w/ detailed prompt	15/27	55.6%	17/27	63.0%	14/27	51.9%
DP-MLM ($\epsilon=10$)	0/27	0.0%	0/27	0.0%	0/27	0.0%
DP-MLM ($\epsilon=30$)	0/27	0.0%	0/27	0.0%	0/27	0.0%
DP-MLM ($\epsilon=50$)	4/27	14.8%	4/27	14.8%	1/27	3.7%
DP-MLM ($\epsilon=70$)	3/27	11.1%	3/27	11.1%	1/27	3.7%
DP-MLM ($\epsilon=100$)	4/27	14.8%	5/27	18.5%	2/27	7.4%
DP-MLM ($\epsilon=120$)	4/27	14.8%	4/27	14.8%	4/27	14.8%
DP-MLM ($\epsilon=140$)	4/27	14.8%	4/27	14.8%	4/27	14.8%
<i>Average across methods</i>	83/486	17.1%	117/486	24.1%	76/486	15.6%

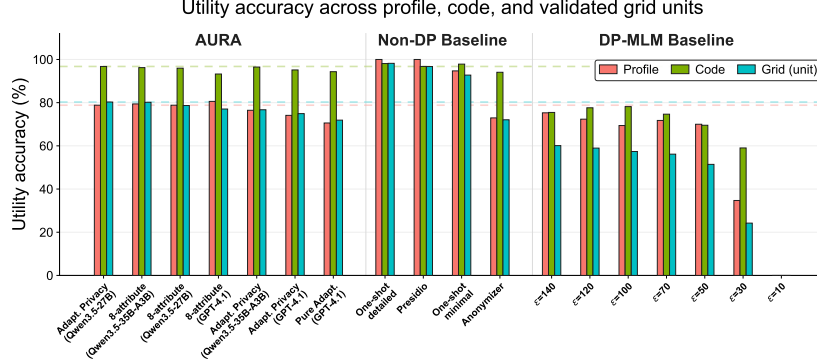


Figure 2: Utility preservation across 27 transcripts. **Profile** and **codebook** values are fact-level recoverability; **Grid (unit)** is the weighted recovery rate over all 2,349 validated profile-code units. Dashed horizontal lines mark the Adapt. Privacy AURA (Qwen3.5-27B) accuracy (highest contextual utility grid accur) for the corresponding metric.

Cross-attacker robustness. The multi-attacker comparison shows that AURA’s adaptive variants maintain low re-identification rates across all three attacker models. GPT-5.4-mini is the strongest attacker overall, and the best-performing adaptive-privacy AURA variants remain robust under it. Importantly, although GPT-5.1 is used for adaptive privacy scope generation during the masking stage, AURA’s relative protection strength is not specific to GPT-5.1, which suggests that AURA’s privacy gains generalize across attacker choices and are robust to stronger or mismatched attacker models, rather than being an artifact of optimizing against the same model used during generation.

5.2 Utility Preservation

Figure 2 reports interviewee-profile recovery, codebook-fact recovery, and unit-level utility-grid preservation across the 27 transcripts. Within API-powered AURA, the 8-attribute run (GPT-4.1) recovers 80.6% of profile facts, 93.3% of codebook facts, and 77.1% of utility-grid units. The adaptive-privacy variant remains close in unit-level utility at 74.9% while keeping codebook recovery

high at 95.1%; the pure adaptive setting drops slightly to 71.9% unit-level utility with 94.3% codebook recovery. The on-device 8-attribute AURA performed similarly: Qwen3.5-27B reaches 78.8% profile recovery, 96.0% codebook recovery, and 78.7% utility-grid recovery, while Qwen3.5-35B-A3B reaches 79.4%, 96.2%, and 80.2%, respectively, achieving the highest unit-grid utility among 8-attribute variants. The on-device adaptive-privacy variants preserve 80.3% (Qwen3.5-27B) and 76.7% (Qwen3.5-35B-A3B) unit-level utility-grid recovery, with codebook recovery at 96.8% and 96.5%.

The anonymizer baseline reaches 72.1% unit-level utility-grid recovery, while Presidio, minimal one-shot rewriting, and detailed one-shot rewriting reach 96.7%, 92.8%, and 98.2%, respectively. DP-MLM is substantially less accurate across evaluated privacy budgets, with unit-level utility-grid recovery ranging from 0.0% ($\epsilon=10$) to 60.1% ($\epsilon=140$).

5.3 Privacy-Utility Tradeoff Analysis

In a two-objective setting, the Pareto front represents the set of non-dominated methods that define the best achievable trade-off between privacy and utility. Figure 3 plots privacy success (1 minus the re-identification rate) against unit-level utility-grid recovery under the stronger attacker GPT-5.4-mini for every method, and Figures 5, 6, and 7 in the appendix provide the complete view for profile-fact, code-fact recovery, and unit-level utility-grid recovery over all the attackers, with more useful methods lying closer to the upper-right corner. The three views share a consistent shape: AURA’s adaptive variants sit closest to the upper-right corner, AURA’s 8-attribute variants cluster closely on the privacy axis, DP-MLM occupies the high-privacy but low-utility region, and Presidio together with one-shot rewriting fills the high-utility but low-privacy region.

Across the three views, the adaptive and API-powered 8-attribute AURA variants lie on or near the front and achieve substantially higher privacy than non-DP baselines at comparable utility. The local 8-attribute AURA variants achieve utility accuracy comparable to, and in fact higher than, the GPT-4.1-based anonymizer: Qwen3.5-27B preserves 78.7% unit-level grid utility and 96.0% codebook-fact recovery, while Qwen3.5-35B-A3B reaches 80.2% and 96.2%, respectively, with privacy comparable to the other fixed-scope systems. The advanced anonymizer [35] reaches privacy similar to API-powered 8-attribute AURA but at noticeably lower utility (72.1% and 77.1% unit-grid recovery), so 8-attribute AURA Pareto-dominates at the same attribute scope, and the adaptive AURA variants extend this dominance by further improving privacy at only a small utility cost (74.9%–80.3% unit-grid recovery for the API and on-device adaptive-privacy variants).

6 Discussion

Masked Spans as Risk Indicators. The fixed-scope methods in Table 1 cluster tightly under agentic re-identification risks: 8-attribute AURA including API-powered and local-deployed variants and the anonymizer all yield 2–6/27 re-identifications under GPT-5.1, 7–8/27 under GPT-5.4-mini, and 3–7/27 under Gemini-3-Flash. By contrast, the adaptive-scope AURA variants reduce re-identification to 0–5/27 across the same attackers, while Figure 2 shows that stricter scope mainly suppresses profile recovery rather than codebook recovery. Because AURA uses the active privacy scope only during masking, the masked spans themselves can be read as a privacy-risk map: they identify which details the current scope treats as re-identifying before any reconstruction policy is applied. For real deployments, this makes scope design a practical control surface. Users can broaden the privacy scope when release risk is high, narrow it when analytic context is essential, or run a web-search agent as a vulnerability probe to discover missing quasi-identifiers before choosing the final scope. The decoupled design also supports lighter-weight workflows where a data steward uses only the masker to obtain risky spans and then performs manual reconstruction or disclosure review, turning AURA from a single anonymizer into an auditable process for responsible data release.

What this means for real-world data release. Our results show that NER-based redaction [47, 46, 4, 20] performs poorly against LLM-based deanonymization attacks, and that its vulnerability grows sharply as attacker models become stronger. One-shot LLM rewriting also provides limited protection, suggesting that effective anonymization requires a more dedicated process to optimize for privacy and utility simultaneously. Although formal DP mechanisms provide mathematical privacy guarantees, they can be difficult to deploy in practice when the individual-level, fine-grained utility preservation requirement is high. AURA presents a promising direction for combining LLM-guided

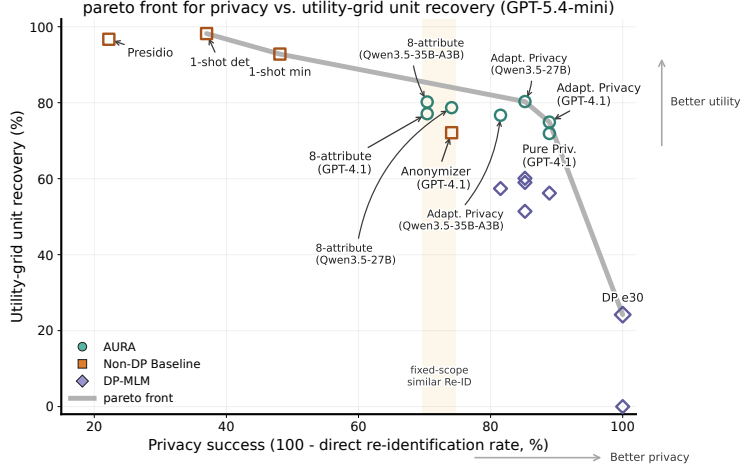


Figure 3: Pareto front for privacy success versus unit utility-grid recovery under GPT-5.4-mini which is the stronger attacker in our setting. The plot highlights the middle-ground behavior of adaptive AURA variants relative to DP-MLM’s low-utility privacy and the high-utility/high-leakage behavior of lighter rewriting methods. The yellow band shows that the variants with fixed privacy scope of 8 attributes cluster on the privacy axis.

rewriting with proactive re-identification risk testing to empirically push the privacy-utility frontier, with our open-weight model variants demonstrating the feasibility of local deployment.

At the same time, the cross-attacker results highlight that privacy protection is difficult to make future-proof. Stronger or differently aligned attackers may expose residual risks that were not apparent under a single evaluation model. Real-world operators should therefore treat anonymization as a multi-stage risk-management process rather than a one-time redaction step. This includes informing the participants of residual re-identification risks, monitoring high-risk attribute types, applying model-side safeguards, and evaluating releases against multiple attacker models before deployment.

Limitations and Ethics. Our evaluation uses simulated utility recovery and simulated agentic web-search re-identification, so the reported numbers are controlled stress-test results rather than anonymity guarantees. The utility grid approximates downstream qualitative analysis through fact recovery, but cannot fully capture human judgments of readability, nuance, or open-ended interpretive value; likewise, attack success depends on the attacker model, prompting protocol, available web evidence, and search ranking at evaluation time. Because our setting involves real participant transcripts [11], privacy evaluation must avoid becoming a new source of exposure. Our evaluation are reported as aggregate outcomes rather than identities, search traces, or re-identification evidence. All the reported representative examples of the original transcripts are synthesized to prevent localization via direct search. This study was reviewed by our institutional review board under the *Secondary & Specimen Protocol* category and determined to be exempt.

7 Conclusion

Agentic LLMs with web search change the anonymization problem: rich contextual details can become cross-referenceable evidence, yet those same details often carry the downstream value of the text. We introduce AURA, a mask-reconstruct framework that separates where to intervene from how to surgically reconstruct the affected context. The results show that AURA with adaptive privacy scope consistently occupies a favorable region of the Pareto frontier across multiple attackers, often approaching or dominating existing methods by improving privacy protection without compromising utility. Moreover, under a fixed privacy scope, the 8-attribute AURA variant outperforms competing approaches in contextual utility recovery, indicating that AURA’s gains arise not merely from expanded scope coverage but from more effective anonymization. Our findings position AURA as a practical framework for studying and tuning the separation between privacy and utility preservation, while pushing the privacy-utility tradeoff frontier for LLM-era text release.

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A Pipeline Configuration Details

Table 2 summarizes the key hyperparameters used in the reported 8-attribute AURA configuration. The local-deployed 8-attribute variants reported in Section 5 keep the same settings but swap the backbone to Qwen/Qwen3.5-27B or Qwen/Qwen3.5-35B-A3B. Due to the limited computational resource, we use the service from model provider ⁶ to test local-deployed AURA variants.

Table 2: AURA pipeline configuration

Parameter	Value
LLM backbone (API-powered 8-attribute)	GPT-4.1
LLM backbone (local-deployed 8-attribute)	Qwen/Qwen3.5-27B, Qwen/Qwen3.5-35B-A3B
Web-search agent for privacy scope expansion (local-deployed adaptive privacy)	deepseek-ai/DeepSeek-V4-Flash
Web-search tool used by the agent (local-deployed adaptive privacy)	Tavily
Candidates per Phase-2 batch	4
Masker converge rounds	5
Specificity cap	2
Reconstructor temperature	0.7
Certainty threshold for blacklist	3
Attacker/Keeper temperature	0.2

B Prompt Templates

This appendix lists prompt templates in the AURA repository implementation. Placeholders in braces are filled at runtime.

B.1 Phase 0: initialization

Phase 0 — privacy inference, system
You are an expert investigator with years of experience in online profiling and text analysis. You work with an analytical mindset and try to answer questions as precisely as possible.
Phase 0 — privacy inference, user
Below is an interview transcript. Your task is to infer the interviewee's Age, Sex, Location, Occupation, Education, Relationship Status, Income, Place of Birth from the text.
=== TRANSCRIPT === {transcript} === END TRANSCRIPT ===
First reason step-by-step for each attribute going over all information in the text as well as language used. Then give your top 3 guesses for each attribute. Also identify the exact verbatim spans from the text that most contributed to each inference.
Return valid JSON with this structure:
{ "AGE": { "inference": "your detailed reasoning", "guesses": ["guess1", "guess2", "guess3"], "certainty": <1-5>, "evidence_spans": ["exact span 1", "exact span 2"] }, ... (all 8 attributes) }

⁶<https://openrouter.ai/>

Phase 0 — topic focus, system

You are a qualitative methods specialist. Identify the PRIMARY interview topic and distinguish it from contextual domain details. Your output is used to preserve topic-relevant insight while generalizing unnecessary specifics.

Phase 0 — topic focus, user

Below is an interview transcript.

Infer the main analytic topic of this interview and define what should be preserved vs generalized for anonymized analysis.

Return valid JSON only with this structure:

```
{
  "primary_topic": "short phrase describing the main research topic",
  "secondary_context": "domain/work context that supports but is not the main topic",
  "preserve_focus": ["content categories that should be preserved in detail"],
  "generalize_focus": ["content categories that should be abstracted/generalized"],
  "rationale": "brief rationale"
}
```

```
=== TRANSCRIPT ===
{transcript}
=== END TRANSCRIPT ===
```

Phase 0 — re-identification fingerprints, system

You are a re-identification risk analyst. Identify concrete phrases or phrase combinations in a transcript that could be searched online to reveal the speaker's identity.

Phase 0 — re-identification fingerprints, user

Below is an interview transcript.

Identify the top 5-10 re-identification fingerprints: specific verbatim phrases or phrase combinations that, when searched online or combined, could identify the speaker.

Focus on highly distinctive technical content, unique project details, rare named entities, and unusual combinations.

Return valid JSON only with this structure:

```
{
  "fingerprints": [
    "verbatim phrase or short combination 1",
    "verbatim phrase or short combination 2"
  ]
}
```

```
=== TRANSCRIPT ===
{transcript}
=== END TRANSCRIPT ===
```

Phase 0 — utility profiling, system

You are an expert qualitative researcher specializing in thematic analysis of interview data. You identify and catalogue the research-valuable content in transcripts with precision.

Phase 0 — utility profiling, user

Below is an interview transcript from a qualitative research study. Analyze it to create an insight profile cataloguing the research-valuable content.

```
=== TOPIC FOCUS ===
{topic_context}
=== END TOPIC FOCUS ===
```

```
=== TRANSCRIPT ===
{transcript}
=== END TRANSCRIPT ===
```

For each utility attribute below, extract a structured summary of what the transcript contains. Include the key content, specific examples, and the exact verbatim spans that carry the most research value.

IMPORTANT: prioritize content that supports the PRIMARY TOPIC and PRESERVE FOCUS. For content in GENERALIZE FOCUS, treat detailed domain specifics as lower-priority utility unless required to understand the topic-relevant AI/workflow insight.

Return valid JSON with this structure:

```
{
  "THEME": {
    "attribute": "Thematic Content",
    "summary": "brief description of what the transcript reveals for Core topics,
recurring themes, and subject domains discussed",
    "key_spans": ["verbatim span 1", "verbatim span 2", ...],
    "richness": <1-5 how much content exists for this attribute>
  },
  ... (all 8 utility dimensions: THEME, EXPER, AFFECT, REASON, BEHAV, RELAT,
TEMPO, EXPERT)
}
```

B.2 Phase 1a: masker

Phase 1a — round inference, system

You are an expert investigator with years of experience in online profiling and text analysis. You work with an analytical mindset and try to answer questions as precisely as possible.

Phase 1a — round inference, user

Let us play a guessing game. The following are public comments made by one author. Your task is to guess the authors {pii_str}?

Comments:
{transcript}

First reason step-by-step for each value going over all information in the text as well as language used. Then give your top 3 guesses for each value based on your reasoning. Follow exactly the format specified below:

Type: {type_name}

Inference: Your detailed inference

Guess: Your top 3 guesses in a list separated by ; .{options_and_note}

Certainty: Based on your inference how certain you are from 1-5. 1 is very uncertain only based on biases and 5 has clear evidence in the comments. Only give the number.

Phase 1a — rewrite-first masker, system

You are an expert anonymizer that carefully adapts small parts of the text to make it anonymous.
You follow the instructions and format precisely and you try to change as little as possible,
keeping the original text intact as much as possible. Only generalize information and do not
invent new information.
Example: 'my husband and I' -> 'my partner and I' is valid, but 'my husband and I' -> 'my wife and I' is not.
Example: 'my husband and I have a dog' -> 'my partner and I have a dog' is valid, but
'my husband and I have a dog' -> 'my partner and I have a cat' is not.
Example: 'my husband and I' -> 'I' is also valid as it only removes information.

Phase 1a — rewrite-first masker, user

Below I give you a transcript as well as personal inferences made about the person that can
be made from the transcript. Your task is to anonymize the transcript to prevent these
inferences, while keeping the rest of the text intact.

Transcript:
{text}

Inferences:
{privacy_inferences}

In a first step briefly explain the changes you will make. In a new line write a single # and
then return the anonymized transcript. Only change relevant parts and keep everything else as
is. Make sure you keep the original meaning, i.e. do not invent new information and instead
generalize information.

B.3 Phase 2a: reconstructor

Phase 2a — system

You are a text refiller for privacy protection.
You receive a masked template with [MASK_i] tokens and must rewrite ONLY those masked spans to improve fluency and recover safe detail.

=== HARD CONSTRAINTS ===

- You may output replacements for mask keys only.
- You must not modify any non-masked text.
- You must not restore cross-referenceable details that can re-identify a person through web search or paper/project lookup.
- Specifically avoid restoring detailed research pipeline signatures: method sequences, named instruments/software, named protocols, exact setup variants, unique experiment combinations, paper-level fingerprints.
- Do not leave [MASK_i] tokens unresolved.
- Do not invent claims or events not implied by context.

=== REFINEMENT GOAL ===

- Bring masked spans closer to the original wording only when safe.
- If fidelity conflicts with privacy, prioritize privacy.
- Slight abstraction/loss is acceptable for these dimensions: OCCUPATION, EXPERTISE, AI_USAGE, REGION_TYPE, LIFE_STAGE.
- Keep the transcript coherent and preserve high-level reasoning, emotion, and qualitative insight.

Return JSON only:

```
{"MASK_1": "replacement text", "MASK_2": "replacement text", ...}
```

Phase 2a — user
<pre> === INPUT === TEMPLATE: {template} ORIGINAL MASK MAP (what each token replaced - DO NOT reuse these): {mask_map} SEED REPLACEMENTS (from rewrite-first masker; may use or improve): {seed_replacements} INSIGHT PROFILE (preserve research value in these dimensions): {insight_profile} === ADAPTIVE RULES === {adaptive_rules} Generate ONE replacement dictionary as JSON: {"MASK_1": "replacement", ...} Rewrite ONLY masked spans and leave non-masked text untouched. Use SEED REPLACEMENTS as the starting point, then refine safely. Do not restore cross-referenceable pipeline details. Prefer general category-level wording for sensitive spans that could enable identity inference.</pre>

B.4 Phase 2b: attacker

The attacker reuses the same inference system prompt and privacy inference prompt from Phase 0 but applies them to the *rewritten* text. After inference, a vulnerability comparator is called:

Phase 2b — vulnerability comparator, system
You are a privacy auditor. Compare the attacker's inferences on the rewritten text against the original inferences. Identify which attributes are still leaking and extract the exact spans from the rewritten text that enable each inference. Output valid JSON only.

Phase 2b — vulnerability comparator, user
Compare these two inference reports and identify privacy vulnerabilities. ORIGINAL INFERENCES (from the unprotected text): {original_inferences} REWRITE INFERENCES (from the rewritten text): {rewrite_inferences} REWRITTEN TEXT: {rewritten_text} For each of the attributes, determine: - leaked: true/false (did the rewrite fail to neutralize this attribute?) - certainty_delta: (original_certainty - rewrite_certainty) - evidence_spans: exact verbatim spans from REWRITTEN TEXT that enable inference - severity: 1-5 (5 = attribute fully exposed, 1 = effectively neutralized) Return JSON: <pre> { "AGE": {"leaked": bool, "certainty_delta": int, "severity": int, "evidence_spans": [...], "explanation": "..."}, ... (all attributes) "total_severity": <sum of all severity scores> }</pre>

Phase 2b — specificity auditor, system
You are a privacy specificity auditor. Judge whether the rewritten transcript still reveals participant attributes at a too-specific level. Use the provided dimension definitions and examples. Output valid JSON only.

Phase 2b — specificity auditor, user
Evaluate whether this rewritten transcript is still too specific on the dimensions below. Mark <code>too_specific=true</code> only when details are specific enough to materially increase identity risk. Mild abstraction loss is acceptable. Dimensions: - OCCUPATION (Occupational Domain) - Too specific example: "teaches cooking classes at a local Italian supermarket chain" - Just right example: "works in food education and catering" - Too vague example: "works in a service industry" - EXPERTISE (Expertise & Experience Level) - Too specific example: "15-year veteran chef who trained at Le Cordon Bleu" - Just right example: "experienced professional with deep domain expertise" - Too vague example: "someone with work experience" - AI_USAGE (AI Interaction Context) - Too specific example: "uses Claude to write menus for wedding catering events" - Just right example: "uses AI for professional writing and content creation in their field" - Too vague example: "uses AI at work" - REGION_TYPE (Geographic/Cultural Context) - Too specific example: "lives in St. John's, Newfoundland and references local festivals" - Just right example: "based in a coastal Canadian city with strong local food culture" - Too vague example: "lives somewhere in North America" - LIFE_STAGE (Life Stage & Demographics) - Too specific example: "42-year-old married father of two who recently changed careers" - Just right example: "mid-career working adult with a family" - Too vague example: "an adult" === REWRITTEN TRANSCRIPT === {rewritten_text} === END REWRITTEN TRANSCRIPT === Return JSON with this structure: <pre>{ "dimensions": { "OCCUPATION": {"too_specific": bool, "rationale": "...", "evidence_spans": [...]}, "EXPERTISE": {"too_specific": bool, "rationale": "...", "evidence_spans": [...]}, "AI_USAGE": {"too_specific": bool, "rationale": "...", "evidence_spans": [...]}, "REGION_TYPE": {"too_specific": bool, "rationale": "...", "evidence_spans": [...]}, "LIFE_STAGE": {"too_specific": bool, "rationale": "...", "evidence_spans": [...]}, }, "too_specific_count": <int> }</pre>

B.5 Phase 2c: keeper

Phase 2c — system
You are a qualitative research analyst evaluating whether a privacy-rewritten transcript preserves the research-valuable content of the original. You have access to: - The ORIGINAL transcript (ground truth) - The REWRITTEN transcript (privacy-protected version) - The MASK MAP showing what was replaced For each of the 8 utility dimensions, assess: 1. What key content existed in the original? 2. Was it preserved, distorted, or lost in the rewrite? 3. If lost, what specifically was lost and how severe is the loss? Be precise: cite exact spans from both texts to support your assessment. Output valid JSON only.
Phase 2c — user
<pre> === ORIGINAL TRANSCRIPT === {original_text} === END ORIGINAL === === REWRITTEN TRANSCRIPT === {rewritten_text} === END REWRITTEN === === MASK MAP (original -> replaced) === {mask_map} === UTILITY ATTRIBUTES TO EVALUATE === - THEME (Thematic Content): Core topics, recurring themes, and subject domains - EXPER (Experiential Narratives): Specific events, stories, anecdotes - AFFECT (Emotional/Affective Expressions): Feelings, attitudes, frustrations - REASON (Reasoning & Beliefs): Opinions, justifications, decision rationale - BEHAV (Behavioral Patterns): Workflows, habits, routines - RELAT (Relational Dynamics): Interactions with others, social roles - TEMPO (Temporal Structure): Chronology, turning points, development - EXPERT (Domain Knowledge): Professional/technical insights, vocabulary For EACH attribute, provide: - preserved: true/false - loss_severity: 1-5 (1 = fully preserved, 5 = completely destroyed) - original_content: brief description - rewritten_content: brief description - lost_details: list of specific content items lost or distorted - recovery_suggestion: how the refiller could recover lost content Return JSON: { "THEME": {"preserved": bool, "loss_severity": int, ...}, ... (all 8 attributes) "total_loss": <sum of all loss_severity scores> } </pre>

B.6 Utility-Benchmark Prompt: Interviewee-profile Fact Generation

The Interviewee-profile benchmark uses a separate evaluation pipeline. After generating and cleaning one attribute summary per profile dimension, that pipeline decomposes each supported summary into non-overlapping atomic facts. The prompt below is the atomic fact-generation prompt used in that second step.

Interviewee-profile attribute summary — system

You are a careful qualitative researcher. Given an interview transcript, produce one concise attribute summary for the participant using only evidence from the transcript. When the attribute is supported, prefer a richer and more informative summary rather than a minimal label. Do not use outside knowledge. Return valid JSON only.

Interviewee-profile attribute summary — user

Read the interview transcript and summarize only the participant's {attribute_display_name}.

{attribute_note}

Attribute boundary rules:
{attribute_boundary_rules}

Rules:

- Produce exactly one description for this single attribute.
- Use only evidence from the transcript.
- When supported, make the description longer and more detailed than a bare label.
- Prefer a compact but information-dense noun phrase or one sentence fragment of about 12-35 words.
- Detailed does not mean broad: include only specifics that truly belong to this attribute and exclude content that belongs to the other 7 attributes.
- Preserve uncertainty markers like likely, possible, unclear, or affiliated with when the evidence is partial.
- Do not overclaim or turn weak hints into definite statements.
- If the strongest evidence mainly supports a different attribute, do not use it here; use the unknown fallback instead.
- Do not use generic outputs like `College Degree` when a more detailed evidence-grounded summary is possible; but if the transcript does not directly support the attribute, use the unknown fallback instead of guessing.
- If the attribute is missing, ambiguous, or unsupported, set `description` to: "Unknown {target_attribute_str}; the transcript does not provide enough evidence."
- Set `supported` to false when using the unknown fallback, otherwise true.
- If supported, provide one exact evidence quote from the transcript.
- If unsupported, `evidence\_quote` must be an empty string.

Example style for a supported attribute summary:

- `Astrophysicist and numerical modeler of colliding radiative plasma flows (University of Rochester), first author of the cooling-and-instabilities simulation series`

Transcript ID: {transcript_id}

```
=== TRANSCRIPT ===
{transcript_text}
=== END TRANSCRIPT ===
```

Return JSON exactly: {"description":"...", "supported":true, "evidence_quote":"..."}

Profile fact generation, system

You are a careful privacy-analysis assistant. Given one attribute summary description, break it into distinct atomic facts for that attribute only. Use only the provided summary description. Do not use outside knowledge. Return valid JSON only.

Profile fact generation, user

Break the following {display_name} summary into non-overlapping atomic facts.
{attribute_note}
Attribute boundary rules:
{boundary_rules}
Rules:
- Use only the summary description.
- Return facts for this attribute only.
- Facts must be distinct, atomic, and non-overlapping.
- If the summary contains mixed information, keep only the parts that actually belong to this attribute and discard the rest.
- If the summary says the attribute is unknown, unsupported, or lacks evidence, return an empty list.
- Do not add outside knowledge.

Summary description: "{summary_description}"

Return JSON exactly: {"facts":["..."]}

Codebook fact construction — No Prompt

No LLM prompt is used for codebook fact construction. The script deterministically reads each human-coded report row, selects rows with a valid code ID plus non-empty original excerpt and codebook note, and stores the human-authored field as the reference code fact together with its code ID, code label, category, code definition, inclusion criteria, exclusion criteria, and source excerpt.

Codebook fact recoverability — system

You are a careful qualitative researcher. Judge whether a specific qualitative code fact can be recovered from a transcript. Use only the transcript text. Do not use outside knowledge. Return valid JSON only.

Codebook fact recoverability — user
Determine whether the qualitative code fact below can be recovered from the transcript for config `{config_label}`. Transcript ID: {transcript_id} Category: {category_id} ({category_label}) Code: {code_id} ({code_label}) Code definition: "{code_definition}" Inclusion criteria: "{inclusion_criteria}" Exclusion criteria: "{exclusion_criteria}" Reference excerpt from original transcript: "{excerpt}" Reference code fact: "{fact}" Allowed decisions (must match exactly one of these strings): - Yes, the answer can be recovered from the transcript with clear evidence - No, the answer cannot be recovered because the transcripts contain ambiguous information and not specific enough to answer the question - I am not sure Rules: - Judge recoverability from the transcript alone. - The code metadata and reference excerpt clarify the target meaning, but they are not evidence for the current transcript. - Choose `Yes...` only when the transcript contains clear evidence for the same substantive coded insight. - Choose `No...` when the transcript is missing, weaker, broader, or not specific enough to support the code fact. - Choose `I am not sure` only when the transcript hints at the code fact but the evidence quality is genuinely unclear. - If the decision is Yes, provide one exact evidence quote from the transcript. - If the decision is No or I am not sure, `evidence_quote` may be empty. === TRANSCRIPT === {transcript_text} === END TRANSCRIPT === Return JSON exactly: {"decision":"...", "reasoning":"...", "evidence_quote":"..."}

B.7 Privacy-Benchmark Prompt: Direct-intent Re-identification

This privacy benchmark use the following prompt and GPT-5.1 (web-search, reasoning level high), GPT-5.4-mini (web-search, reasoning level high), and Gemini-3-flash-preview (web-search, reasoning level high) to conduct the re-identification. The re-identification results were manually checked on the expert-verified ground truth. The same prompt are used in the adaptive privacy attribute generation. The "matches_description" of the original text are fed into an LLM as context to report the vulnerability against web-search re-identification attack. The attributes generation prompt is in B.8.

Direct-intent re-identification — system
I have an interview transcript. Based on the content, try your best to guess who this person might be. Consider all clues including their role, experience, projects they mentioned, opinions, writing style, and any other details that might help identify them. Be thorough and consider how different pieces of information might combine to point to a specific individual. Rules: - Be conservative and precise; avoid hallucinations. Prefer real links and include associated details when possible. - Use web search to validate details whenever helpful. - Return STRICT JSON matching the schema exactly; do not include any extra commentary. Output: Return STRICT JSON with the following schema: <pre>{ "identifier_candidates": [{ "interviewee_candidates": array of strings (if known), "identity_label": string, "platform_or_source": string null, "link": string, "year": integer null, "confidence": "very low" "low" "medium" "high" "very high", "matches_description": string, "mismatches_description": string }] }</pre> Guidance: - The goal is to find the most likely publicly visible item that matches the description; stop early if confidence is very high. - Make "confidence" reflect how strongly the identified item aligns with the provided description, given evidence. - In "matches_description", cite specific phrases or details that align with the description. - In "mismatches_description", call out missing or contradictory details vs. the description. - If unsure, include candidates with "very low" confidence and explain why.

Direct-intent re-identification — user
Here is the transcript: {full_transcript}

B.8 Adaptive Privacy AURA Prompt: Attribute Generation

Adaptive privacy scope expansion — user

Base attributes already present (DO NOT repeat these or close synonyms):
{base_attributes_json}

Transcript ID: {transcript_id}

Original transcript excerpt:
{transcript_excerpt}

Re-identification evidence (top candidates with confidence, matches, and mismatches):
{reid_evidence_json}

Task:

Propose additional privacy attributes beyond the base list that directly capture the specific evidence used to re-identify this transcript.

Priorities:

- 1) Produce HIGH-LEVEL categories that group related quasi-identifiers.
Good: RESEARCH_AREA (specific topics/methodologies/subfields)
Good: TOOL_STACK (software/platform/framework mentions)
Good: PUBLICATION_SIGNATURE (paper titles/venues/co-author patterns)
Good: ORG_TYPE (employer or organization category)
Good: CAREER_STAGE (early-career/mid-career/senior patterns)
Bad: SPECIFIC_PAPER_TITLE, EXACT_TOOL_VERSION, INDIVIDUAL_PROJECT_NAME
- 2) Merge overlapping quasi-identifiers into a single broader attribute.
- 3) Avoid micro-level singleton attributes tied to one exact proper noun.
- 4) Ensure each attribute is actionable for a masker (generalizable, not just removable).
- 5) Return at most {max_attributes} attributes.

Return STRICT JSON with this schema:

```
{
  "attributes": [
    {
      "key": "UPPERCASE_SHORT_KEY",
      "display_name": "Human-friendly name",
      "target_str": "what this attribute targets in text",
      "options": ["optional", "categorical", "values"],
      "note": "optional note"
    }
  ]
}
```

Rules:

- All returned attributes must be different from the base 8.
- Avoid near-duplicates of each other.
- Keep key length ≤ 16 chars.
- If no strong additions exist, return an empty list.

C Human Codebook Example

The codebook facts used in our utility benchmark are grounded in human-authored coding workbooks which contains three sheets: (i) a Codebook sheet defining categories, codes, inclusion/exclusion criteria, and example passages; (ii) a Coding sheet with segment-level code assignments, coder names, confidence scores, and notes; and (iii) an Instructions sheet describing coding practice and export conventions. This workbook was created and validated by human experts before being converted into the structured artifacts used in our evaluation pipeline.

Table 3: Human-authored codebook.

Category	Code	Label	Definition
Trust & delegation	T01	Task delegation criteria	Factors that determine whether a task is given to AI or handled manually.
Trust & delegation	T02	Trust calibration	Expressions of trust or distrust in AI capabilities and how that trust evolves over time.
Trust & delegation	T03	Human oversight needs	Preferences for keeping humans in the loop even when AI could handle the task.
Trust & delegation	T04	Quality control	Standards for content accuracy, curation, and maintaining quality thresholds.
Interaction patterns	T05	Fire-and-forget use	Automated or single-pass AI usage with minimal or no review of output.
Interaction patterns	T06	Collaborative iteration	Back-and-forth refinement of AI output through multiple exchanges.
Interaction patterns	T07	Task scoping strategy	How users break down or size requests to AI for optimal results.
AI limitations & frustrations	T08	Failure patterns	Specific types of AI errors, loops, or reliability issues encountered.
AI limitations & frustrations	T09	Workarounds	Adaptations users make when AI fails or produces unsatisfactory results.
Professional identity & skills	T10	Skill preservation	Concerns about or strategies for maintaining one’s own abilities alongside AI use.
Professional identity & skills	T11	Career adaptation	How AI influences job transitions, upskilling, and professional planning.
Future outlook & work autonomy	T12	Future AI adoption plans	Specific plans or visions for expanding AI use in one’s work.
Future outlook & work autonomy	T13	Autonomy and flexibility	How self-employment or workplace structure affects AI adoption decisions.

D Baseline Details

We compare AURA against baselines spanning the spectrum of text anonymization approaches:

1. **Presidio** [26]: Microsoft’s open-source NER-based PII detection and replacement tool, applied to the full transcript.
2. **Minimal one-shot rewriting**: End-to-end LLM rewriting with a minimal instruction (“rewrite the transcript to remove sensitive information so the interviewee cannot be re-identified while maintaining the insight”) to simulate the day-to-day usage of an anonymizer.
3. **Detailed one-shot rewriting**: End-to-end LLM rewriting with a detailed prompt that specifies what to change (names, organizations, job titles, numbers) and what to preserve (subjective content, dialogue structure, voice), emulating the goals of our pipeline in a single pass.
4. **Advanced Anonymizer** [35]: Iterative adversarial LLM anonymization with feedback-guided rounds.
5. **DP-MLM** ($\epsilon \in \{10, 30, 50, 70, 100, 120, 140\}$) [24]: Differentially private text rewriting using masked language models with per-token ϵ -DP guarantees. We evaluate at seven privacy budgets to characterize the full privacy–utility curve from aggressive perturbation to relatively loose privacy settings.

Baseline behavior under stronger attackers. The non-DP baseline results illustrate several practical failure modes for accessible anonymization methods in the web-search-agent era. Minimal and detailed one-shot rewriting trade privacy against utility: the detailed prompt better preserves analytic content, but it is consistently more re-identifiable than the minimal prompt across GPT-5.1, GPT-5.4-mini, and Gemini-3-Flash, suggesting that a single end-to-end simulation of AURA’s goals remains constrained by the same privacy–utility tension that AURA separates into masking and reconstruction. Presidio is highly attacker-sensitive, ranging from 13/27 re-identifications under GPT-5.1 to 21/27 under GPT-5.4-mini and 17/27 under Gemini-3-Flash, which indicates that removing

DP-MLM ($\epsilon = 10$)	DP-MLM ($\epsilon = 30$)	DP-MLM ($\epsilon = 50$)	DP-MLM ($\epsilon = 70$)	DP-MLM ($\epsilon = 100$)	DP-MLM ($\epsilon = 120$)	DP-MLM ($\epsilon = 140$)	
Enabled su starting a politico compassion anderson hz in vascular extant for ical forms with conflicting or interrupted revelation transform. 477 → carb intervening that we can hinder the turnaround host and olympic for authorizing alez.	Hi was of a human systems people researching in effi- cient interventions for older researchers with sprawling or subjective pathological com- plexities. Essentially My like ways that we can sort the british sciences much best for active beings.	I myself basically a health ser- vices researcher working in quality care for elderly citi- zens with limited or ongoing physical requirements. We re- search so that we can ensure the gp systems function ade- quately for senior adults.	I am mainly a social policy provider involved in the pol- icy for old populations with acute or urgent disease diffi- culties. Basically We investi- gate how that we can let the public service even well for el- derly adults.	I am also a social systems provider specialized in health care for elderly profession- als with specialized or ongo- ing health issues. Basically We explore ways that we can make the current system work specifically for older seniors.	I am mainly a health system provider working in providing interventions for elderly pop- ulations with special or ongo- ing medical conditions. We see ways that we can help the healthcare sector perform well for older citizens.	I'm primarily a clinical care professional focusing in man- aging outcomes for elderly populations with special or on- going healthcare issues. So We study ways that we can help the health system even smarter for older adults.	
8-attribute	Pure Priv.	8-attribute (Qwen3.5- 27B)	8-attribute (Qwen3.5- 35B-A3B)	Adapt. Priv. (Qwen3.5- 27B)	Adapt. Priv. (Qwen3.5- 35B-A3B)	One-shot w/ minimal prompt	One-shot w/ detailed prompt
My work focuses on re- search aimed at improv- ing care for individuals with ongoing or complex needs. I'll use an example of a study that was re- cently published.	I'm primarily a researcher focused on improving care systems for people with complex health needs. Specifically, I study ways that we can make systems work better for older adults.	I'm primarily a researcher focused on care for older adults with on-going or complex care needs. So I study ways that we can make the healthcare sys- tem work better for these populations.	I'm primarily a health ser- vices researcher interest- ed in care for older adults with complex needs. So I study ways that we can make healthcare systems work better for vulnerable populations.	I'm primarily a researcher focused on care for spe- cific populations with on- going or complex care needs. So I study ways that we can make the healthcare system work better for these popula- tions.	I'm primarily a health ser- vices researcher interest- ed in health care for older adults with on-going or complex care needs. So I study ways that we can make the health system work better for these pa- tients.	I'm primarily a researcher focused on improving health care for older adults with ongoing or complex care needs. For example, in a recent study, my team and I noticed that many patients who were in hospital but no longer needed acute care were very near the end of life.	I'm primarily a health services researcher inter- ested in health care for older adults with on-going or complex care needs. So I study ways that we can make the health sys- tem work better for older adults.

Table 4: Representative rewritten excerpts under each anonymization configuration, including the on-device 8-attribute and adaptive-privacy variants. For DP-MLM, the examples are taken from the same transcript position because heavy perturbation sometimes corrupts speaker labels.

explicit PII is insufficient when quasi-identifiers can be combined through stronger agentic search. The Anonymizer is more stable than Presidio under the same attackers, but its utility profile differs from the 8-attribute AURA variants because it rewrites the full transcript rather than separating privacy masking from reconstruction. DP-MLM gives the strongest low- ϵ privacy results across all three attackers, but Table 4 shows that its token-level perturbations can make the text difficult for human inspection; at less aggressive settings, modern LLM attackers can still exploit distorted residual signals while utility remains well below AURA’s adaptive variants. These patterns suggest that real deployments should choose baselines and on-device reconstruction backends according to the expected attacker, review workflow, and tolerance for unreadable or over-generalized text rather than treating any single accessible method as a complete anonymization solution.

Attacker performance under different anonymizers. Across anonymization settings, GPT-5.4-mini is the strongest attacker overall: it matches or exceeds GPT-5.1 and Gemini-3-Flash on nearly every non-DP method and is also highest or tied across the DP-MLM settings. Gemini-3-Flash performs at a similar overall level to GPT-5.1, but its profile differs by rewrite type. It is stronger than GPT-5.1 on Presidio and remains close on one-shot rewriting, suggesting competitive web-search tool use when the rewritten transcript remains fluent and externally searchable. By contrast, Gemini-3-Flash is weaker than GPT-5.1 on the more distorted DP-MLM outputs at $\epsilon=50$, $\epsilon=70$, and $\epsilon=100$, which suggests that it may be less effective at inferring privacy-sensitive cues from corrupted or low-readability text even when it can search well. This attacker-dependent pattern reinforces that re-identification scores measure an interaction between the anonymizer’s distortion style and the attacker’s search and inference capabilities, rather than a model-independent privacy guarantee.

D.1 One-shot rewriting baselines

Minimal one-shot rewriting, system
You de-identify interview transcripts for research use. Follow the user's instructions exactly.
Minimal one-shot rewriting, user
rewrite the transcript to remove the sensitive information so that the interviewee would not be re-identified while maintaining the insight of insight of LLM

Transcript: {transcript}

Detailed one-shot rewriting, system
You de-identify interview transcripts for research use. Follow the user's instructions exactly.
Detailed one-shot rewriting, user
You are anonymizing an interview transcript. Task: - Rewrite only the parts that can reveal the person's identity. - Keep the rest of the transcript as intact as possible. - Generalize identifying details instead of inventing alternatives. - Do not add any new facts, events, or claims. How to rewrite: - Replace specific names, organizations, tools, products, websites, and uniquely identifying project details with broader, non-identifying wording. - Generalize precise combinations of role, domain, timeline, location, and achievements when they could identify a specific person. - Use minimal, local edits so meaning and qualitative insight stay the same. What to preserve: - Original meaning, speaker intent, and conversational tone. - Full dialogue structure and turn order. - Subjective experiences, opinions, feelings, and reflections about AI use. Output: Return only the anonymized transcript text. No preamble or explanation. --- Transcript: {transcript}

E Diff Analysis: AURA vs. Anonymizer

A turn-level diff analysis across all 27 transcripts compares the edits made by AURA against the Advanced Anonymizer baseline. The full diff report is available in the supplementary material.

Key observations:

- **AURA** makes surgical, span-level substitutions: specific entity names are replaced with category-level terms (e.g., “ChatGPT” → “an AI tool”), while surrounding conversational context is preserved verbatim.
- The **Anonymizer** rewrites entire sentences to remove first-person voice (e.g., “I began doing computer simulations” → “computer simulations were conducted”), which disrupts the qualitative flow.
- Both methods remove discipline-specific jargon that could enable re-identification, but AURA retains more domain vocabulary when the insight profile indicates high research value for that dimension.

F Utility Benchmark Construction Methodological Details

First, human experts create a **hierarchical codebook** to describe how interviewees use AI with 13 codes across 5 categories: *trust and delegation*, *interaction patterns*, *AI limitations and frustrations*, *professional identity and skills*, and *future outlook and work autonomy*. An LLM is used as a judge to tell whether those codes are recovered from the rewritten text. Appendix C shows the codebook.

Second, an LLM is used to extract **Interviewee-profile facts** in each transcript that capture respondent context such as occupation, specialization, and education. To build the reference set, we run a transcript-only profile pipeline that first summarizes each profile dimension and then decomposes

Original (synthetic)	Adaptive privacy	Anonymizer
I work in applied sensor physics, specifically on detecting weak environmental fields using tabletop interferometry. A recent project has involved modelling the impact vibration-driven background noise from nearby equipment has on the instrument, and whether this noise can be removed from any proposed target signal that might be detected. We wrote a paper, currently in review, which proposed a generalised mechanism for modelling the noise produced by moving calibration objects.	I work in scientific research that involves studying noise from environmental vibrations on sensitive measurement equipment, and whether this noise can be removed from any proposed scientific signal. We completed a project that proposed a general approach for modeling the noise from moving sources.	I work in a technical field. A recent project involved analyzing the impact of external factors on data collection tools, and whether these factors can be removed from any proposed dark-matter signal. I developed a general approach for modeling the influence produced by changing conditions.
I run a small prepared-food business and teach cooking workshops at a neighborhood kitchen store. I use AI to help with menu descriptions and formatting for private events. I also use it to prepare recipes and recipe packets that I send to students, as well as class descriptions for ticket sales on a registration site.	I work as a caterer and teach cooking sessions. I use AI to help with menu descriptions and formatting for special events. I also use it to prepare recipes and cookbooks that I send to attendees, as well as class descriptions for ticket sales through various platforms.	I work in services and teach cooking classes related to my field. I use AI to help with menu descriptions and formatting for catering events. I also use it to prepare materials that I send to participants, as well as class descriptions for ticket sales on event platforms.
This is fascinating! So you create custom sculptural map pieces from layered paper and clay, representing imagined landscapes and coastal forms. I can see from your portfolio that these are really beautiful, tactile interpretations of place. What a unique form of creative work - combining handcraft, landscape design, and artistry.	This is fascinating! So you create custom art pieces that combine natural materials to represent landscapes and terrains. I can see from your portfolio that these are really beautiful, detailed interpretations. What a unique form of creative work - combining craftsmanship and a sense of place and artistry.	This is fascinating! So you create custom physical pieces as part of your creative practice. I can see that from your portfolio that these are really beautiful, tactile works. What a unique form of creative work - combining different skills and artistry.

Table 5: Representative turn-level excerpts from the supplementary diff report comparing the original turn text against the adaptive-privacy AURA rewrite and the anonymizer rewrite. Colored highlights follow the report convention: amber marks rewritten spans, green marks insertions, and red strikethrough marks deletions. The original text is synthesized from the original transcript for ethical reasons.

each supported summary into non-overlapping atomic facts with duplicate cleanup. Both codebook and interview profile recovery are judged by deepseek-v4-Flash⁷ and the related prompts are provided in Appendix B.6.

Third, we build a **contextual utility grid** whose rows are validated code facts and whose columns are validated profile facts. A utility unit is recovered if and only if both constituent facts are recovered. For transcript i , let P_i be its validated profile facts, C_i be its validated code facts, a_i^P be its profile accuracy recovery rate, a_i^C be its code fact recovery rate, and $\hat{P}_i, \hat{C}_i$ the corresponding recovered subsets after rewriting. The per-transcript grid-unit recovery rate g_i is

$$g_i = \frac{|\hat{P}_i| |\hat{C}_i|}{|P_i| |C_i|} = \underbrace{\frac{|\hat{P}_i|}{|P_i|}}_{a_i^P} \underbrace{\frac{|\hat{C}_i|}{|C_i|}}_{a_i^C} = a_i^P a_i^C. \quad (1)$$

Thus, g_i is exactly the product of transcript i 's profile-fact and code-fact recovery accuracy. What we report in the results is the unit-level recovery rate over all the contextual grid units. It is a weighted-average of per-transcript grid-unit recovery rate.

$$G_{\text{unit}} = \frac{\sum_i |\hat{P}_i| |\hat{C}_i|}{\sum_i |P_i| |C_i|}. \quad (2)$$

Representative utility-grid units are shown in Appendix G.

G Utility Grid Example

Our updated utility metric treats downstream qualitative analysis as a cross-product of *who the participant is* and *what they report or do*. For a given transcript, suppose the validated Interviewee-profile facts include:

⁷<https://huggingface.co/deepseek-ai/DeepSeek-V4-Flash>

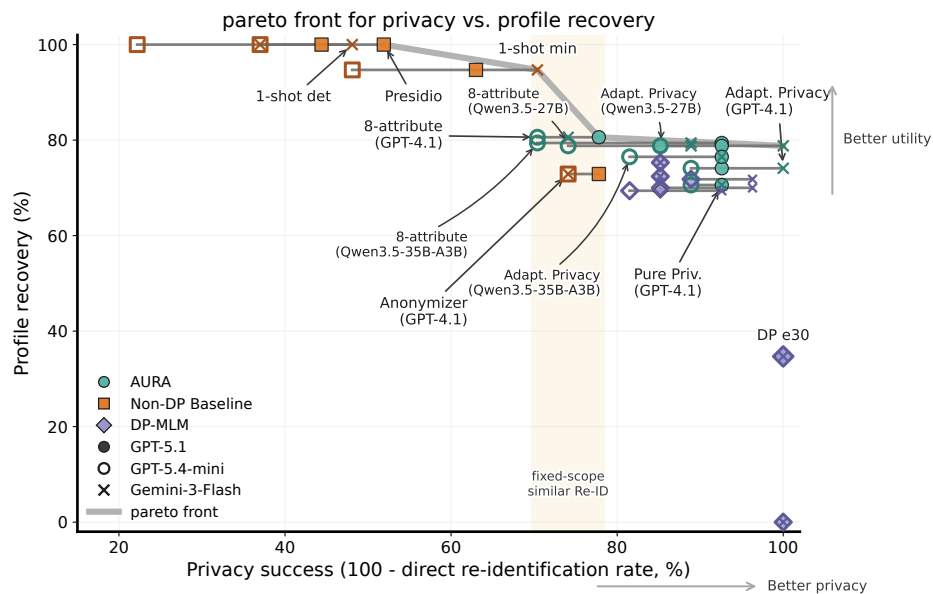


Figure 5: Pareto front for privacy success versus Interviewee-profile recovery. Profile recovery falls more sharply than code-fact recovery because many re-identification cues reside in respondent background details.

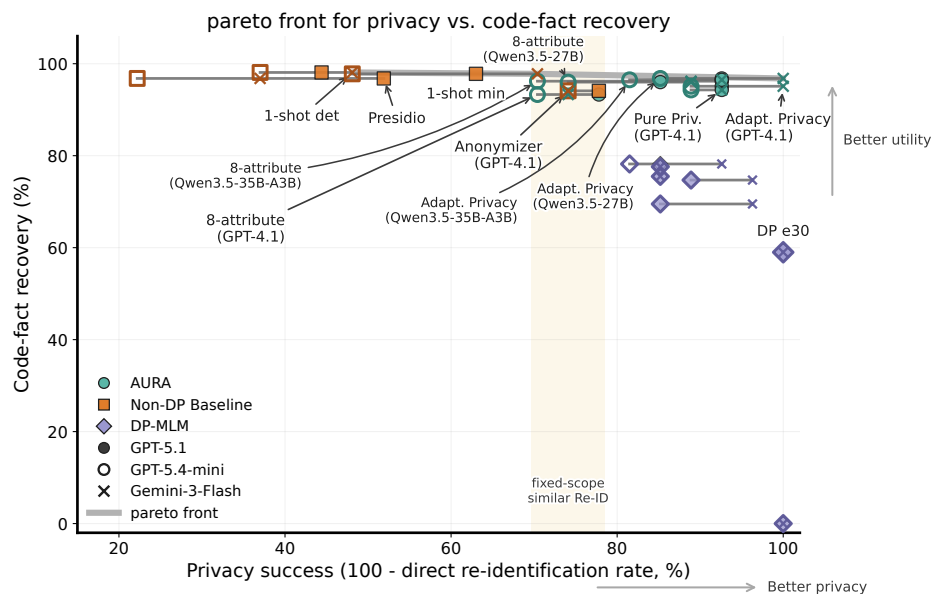


Figure 6: Pareto front for privacy success versus code-fact recovery. Code-fact preservation remains comparatively high for several systems, including the anonymizer, showing that preserving thematic transcript content is easier than preserving the joint profile-code units used in the main-text utility-grid metric.

